

MASTERS PROJECT MILESTONE REPORT

Name of the student:

Name of the supervisor(s):

Title of the project (may be tentative): Poisson Surrogate for High-Dimensional Multinomial Regression Models

In many modern applications, data naturally arise as counts distributed across multiple categories (Taddy, M 2015). Modelling such data requires flexible statistical frameworks that can capture both categorical structure and dependence among observations. Multinomial regression modelling is widely used to analyse count data across multiple categories. In many of these settings, observations are naturally grouped, and dependence among responses can be captured through random effects, leading to multinomial mixed models. However, the likelihood of multinomial mixed models does not admit a closed-form expression except for the special case when there are no covariates. This lack of a tractable likelihood representation creates substantial computational challenges for estimation and inference.

To address this issue, Lee et al. (2017) proposed the so-called “Poisson trick”, which reformulates the multinomial likelihood using a Poisson surrogate. This approach is used in the context of categorical data analysis to fit multinomial regression models by assuming independent Poisson distribution for the counts, which provides a more tractable framework and allows standard likelihood-based inference to be carried out efficiently.

Lee et al.(2017) had focused on low-dimensional datasets, where the number of covariates p is smaller than the sample size n . In contemporary applications, however, one encounters regimes in which $p \gg n$ or where the total parameter dimension grows with both the number of predictors and response categories. In such settings, classical multinomial methodology faces serious difficulties including non-identifiability, instability of maximum likelihood estimation, overfitting and increased computational cost.

Motivated by this perspective, our work will investigate high-dimensional extensions of multinomial modelling based on Poisson surrogate framework. We seek to combine tractable likelihood reformulations with appropriate regularization techniques to achieve both statistical reliability and computational scalability since modern multinomial regression models have more covariates and potentially complex dependence structures.

Student’s signature

Supervisor’s signature

Date: